Fairness Auditing of an Income-Classification Model: SHAP-Based Analysis of Proxy Features

Nguyen Lam Thai
HUST
Thai.NL261068M@sis.hust.edu.vn

Dinh Trong Nghia
HUST
Nghia.DT261074M@sis.hust.edu.vn

Nguyen Trong Tuan
HUST
Tuan.NT261067M@sis.hust.edu.vn

Vuong Dinh Minh
HUST
Minh.VD261073M@sis.hust.edu.vn

Nguyen Tuan Minh
HUST
Minh.NT261078M@sis.hust.edu.vn

Abstract

Removing a sensitive feature from a machine-learning model does not necessarily remove bias when other features encode related information. We study this issue in income classification using the UCI Adult dataset, treating `sex` as the sensitive attribute. We compare XGBoost models trained with and without `sex`, use TreeSHAP and feature–sensitive-attribute association to identify proxies, retrain models after proxy ablation, and evaluate Fairlearn `ExponentiatedGradient` models with demographic-parity and equalized-odds constraints. On the fixed held-out test split, removing `sex` changed predictive performance negligibly and reduced demographic-parity difference from 0.1710 to 0.1662, while equalized-odds difference changed from 0.0604 to 0.0575. Jointly removing the selected candidate proxies reduced demographic-parity difference to 0.1576 and equalized-odds difference to 0.0528. Fairness-constrained learning produced larger changes in the targeted metric: demographic parity reduced demographic-parity difference to 0.0122, whereas equalized odds reduced equalized-odds difference to 0.0409. These interventions also produced different performance and cross-metric trade-offs. The results support a limited conclusion: feature unawareness is not a sufficient fairness audit, and proxy-focused and constraint-based interventions should be evaluated against an explicitly chosen fairness criterion rather than treated as interchangeable definitions of fairness.

1 Introduction

Machine-learning models can produce unequal outcomes across demographic groups even when their average predictive performance is high [Mehrabi et al., 2021]. A common response is to remove a sensitive attribute, such as `sex`, from the model inputs. This intervention is simple and often presented as a way to prevent discrimination. However, other features may still contain information associated with the protected attribute (i.e., `sex`) and allow the model to reproduce group disparities [Datta et al., 2017]. Removing `sex` therefore does not by itself establish that a model is fair.

This paper studies that problem in income classification, where `sex` is treated as a sensitive feature because it marks a socially salient group for which historical and structural inequalities can affect

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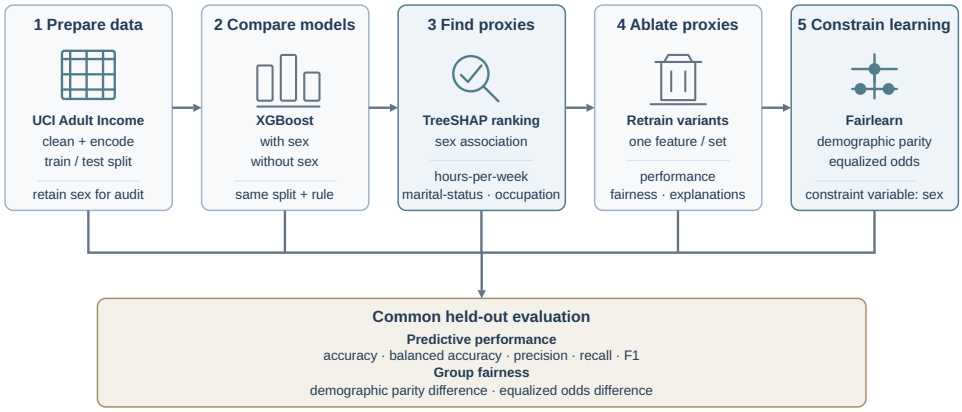


Figure 1: Overview of our method.

access to education, occupations, wages, and other opportunities represented in the data [Mehrabi et al., 2021, Barocas et al., 2019]. The concern is not that sex is inherently inappropriate for prediction or that it should never be observed; rather, using it, or information that acts as its proxy, may allow a model to reproduce or amplify unequal treatment in decisions related to economic opportunity. We use the UCI Adult Income dataset, which contains demographic, educational, occupational, and work-related features and a binary target indicating whether annual income exceeds $50,000 [Dua and Graff, 2019]. The dataset provides a useful setting for a controlled audit because it contains an explicit `sex` attribute alongside several features that may encode social and economic structure.

In this study, we ask whether excluding `sex` changes the fairness of an income-classification model and whether remaining features can act as proxies for it. To answer this question, we (1) compare models trained with and without `sex`, (2) investigate whether selected remaining features contribute to group-related disparities, and (3) compare feature removal with fairness-aware mitigation. Predictive performance and group fairness are evaluated together so that an apparent fairness improvement is not interpreted without considering its effect on utility. An overview of our method is provided in Figure 1. This paper makes three contributions:

- We provide a reproducible comparison of income-classification models trained with and without the explicit sensitive attribute `sex`.
- We combine model explanations and feature ablation to investigate whether specific remaining features behave as potential proxies.
- We compare proxy removal with fairness-constrained learning using common predictive and group-fairness metrics, making the associated performance–fairness trade-offs explicit.

The remainder of this paper is structured as follows. Section 2 discusses related work; section 3 formulates the problem; section 4 describes the method, dataset, and evaluation metrics; section 5 presents the experimental results; section 6 discusses the limitations and future work; finally, section 7 concludes.

2 Related Work

Fairness through unawareness, proxy discrimination, and causal fairness. One influential response to discriminatory prediction is to omit protected attributes from the model. However, Dwork et al. [2012] argue that fairness cannot be reduced to ignoring sensitive attributes: their *Fairness Through Awareness* framework instead requires similar individuals, under a task-specific similarity metric, to receive similar treatment. This distinction is directly relevant to our comparison between models with and without `sex`: removing the column is an unawareness baseline, not evidence that the remaining representation is fair. The proxy discrimination literature makes the same concern operational by studying how protected-group information can be inferred from seemingly permissible

features and how such relationships can be audited [Datta et al., 2017]. By contrast, Kusner et al. [2017] define counterfactual fairness through a causal model, requiring a prediction to remain unchanged in counterfactual worlds in which an individual's protected attribute differs. Our SHAP rankings and ablations are deliberately more modest: they can identify predictive association and model dependence on candidate proxies, but they cannot determine whether a feature is a causal mediator, a legitimate correlate, or an impermissible proxy.

Group fairness criteria and the problem of incompatible objectives. A second body of work formalizes how disparities should be measured once a model has been trained. Hardt et al. [2016] introduce equality of opportunity and equalized odds, connecting fairness to conditional error rates and providing a post-processing procedure that uses group membership to alter predictions. Yet no single criterion resolves the broader normative question. Chouldechova [2017] show that calibration and error-rate parity can conflict when outcome prevalence differs between groups, while Kleinberg et al. [2017] prove closely related impossibility results for risk scores. Critical reviews likewise emphasize that criteria such as statistical parity encode different values and can lead to different conclusions about the same system [Corbett-Davies and Goel, 2017, Barocas et al., 2019]. Accordingly, our audit reports demographic-parity difference and equalized-odds difference as separate descriptive outcomes alongside accuracy, balanced accuracy, precision, recall, and F1. The UCI Adult Income experiment therefore tests whether an intervention changes particular measured disparities; it does not treat a small value for one metric as a complete definition of fairness.

Explainability and feature-level auditing for proxy detection. The third relevant line of work uses explanations to inspect how models reach their predictions. Lundberg and Lee [2017] introduce SHAP, an additive feature-attribution framework grounded in Shapley values whose local accuracy and consistency properties make it attractive for comparing contributions across individual predictions. We use TreeSHAP in this spirit to rank features in the sex-excluded XGBoost model and to prioritize variables such as `relationship`, `marital-status`, `occupation`, and `hours-per-week` for further investigation. SHAP, however, is an explanatory device rather than a fairness test: a large attribution indicates model reliance, not discriminatory intent or causal influence. This caveat is especially important for UCI Adult Income dataset because its features are dependent. Aas et al. [2021] show that common SHAP approximations can be misleading when feature independence assumptions fail and propose dependence-aware approximations and grouped explanations. Our use of explanations is therefore paired with feature ablation. The combination asks a narrower and testable question than explanation alone: whether removing a candidate feature changes both model behavior and observed group disparities on the same held-out population.

Fairness-aware mitigation and constrained learning. Feature deletion is only one possible intervention, and it may remove useful predictive information without directly controlling the disparity of interest. Zafar et al. [2017] incorporate fairness constraints into the geometry of a classifier's decision boundary, making the fairness–accuracy trade-off an explicit part of training. Madras et al. [2018] instead learn representations that make sensitive attributes difficult to predict using an adversary, relating the training objective to demographic parity, equal opportunity, and equalized odds. The reductions approach of Agarwal et al. [2018] is particularly close to our implementation: it converts constrained binary classification into a sequence of cost-sensitive classification problems and can optimize error subject to demographic-parity or equalized-odds constraints. This provides the conceptual basis for Fairlearn's `ExponentiatedGradient` comparison in our audit. Our design places these approaches beside, rather than above, sex removal and proxy ablation: all variants use the same fixed split and held-out metrics, allowing us to distinguish fairness obtained by deleting features from fairness obtained by explicitly constraining predictions. The comparison also preserves `sex` for evaluation and mitigation constraints while excluding it from ordinary predictive features, so that the fairness intervention is measurable without treating the sensitive attribute as a standard model input.

Taken together, prior work establishes four points that motivate our study: (1) omitting a sensitive feature does not rule out proxy discrimination, (2) fairness criteria measure different and sometimes incompatible properties, (3) explanations can reveal model reliance but do not by themselves establish causality or unfairness, and (4) constrained learners offer a direct alternative to feature deletion. The gap addressed here is an integrated, reproducible audit that connects these pieces in one UCI Adult Income experiment: it compares explicit `sex` removal with targeted proxy ablation, uses SHAP and

subgroup diagnostics to justify which proxies are tested, and evaluates both feature interventions and Fairlearn reductions under common predictive and group-fairness metrics.

3 Problem formulation

Let $\mathcal{D} = \{(x_i, a_i, y_i)\}_{i=1}^{n}$ denote the UCI Adult Income dataset, where x_i contains the observed demographic, educational, occupational, and work-related features, $a_i \in \{0, 1\}$ denotes `sex`, and $y_i \in \{0, 1\}$ indicates whether the individual's income exceeds \$50,000. The objective is to learn a binary classifier

$$f_\theta : \mathcal{X} \to \{0, 1\}, \qquad \hat{y}_i = f_\theta(x_i),$$

that predicts the income label from the available features.

We study two feature representations. In the full-feature setting,

$$x_i^+ = (x_i, a_i),$$

where `sex` is included as a model input. In the sex-excluded setting,

$$x_i^- = x_i,$$

where `sex` is withheld from prediction but retained for evaluation. The classifier parameters are learned on the training split by minimizing a classification loss, while predictive performance is assessed on the held-out test split. The central empirical question is whether excluding a_i changes predictive performance or group-level outcomes, and whether other features act as proxies for a_i.

4 Methodology

4.1 Our method

Figure 1 summarizes the fairness audit as a five-stage workflow: prepare the UCI Adult Income dataset, select the classifier and compare XGBoost models with and without `sex`, identify and ablate candidate proxy features, and apply fairness constraints before evaluating every variant on the same held-out test set.

Prepare the data. We first clean and encode the dataset using a deterministic preprocessing pipeline. Preprocessing is fitted on the training data only and then applied to the held-out data to prevent leakage. The sensitive feature, `sex`, is retained outside the model feature matrix so that it remains available for measuring association when it is excluded from model training. Further details about the dataset and preprocessing procedure are given in 4.2.

Select a classifier and compare models with and without `sex`. Before the fairness audit, we compare eight classifiers using the same official training and held-out test partitions and the same training-only preprocessing. The comparison includes Logistic Regression, Decision Tree, Random Forest, HistGradientBoosting, CatBoost, Support Vector Machine, KNN, and XGBoost. We select the model with the strongest overall predictive balance, giving primary consideration to balanced accuracy and F1 because the income labels are imbalanced, while also reporting fairness metrics. XGBoost provides the best balanced accuracy, recall, and F1 in this comparison and is therefore used for the subsequent audit. We then train matched XGBoost models, with one baseline including `sex`, while a paired model removes only that feature. This comparison isolates the effect of explicit feature removal while leaving the model family and experimental conditions unchanged. The sex-excluded model is therefore treated as an unawareness baseline, not as evidence that the remaining features are free of sensitive information.

Identify candidate proxy features. We analyze the sex-excluded model with TreeSHAP and through its feature association with `sex` [Lundberg and Lee, 2017]. Features are treated as candidate proxies when they both contribute to predictions (i.e., are used by the model) and have a high association with `sex` (i.e., encode `sex`).

Table 1: Summary Statistics of the UCI Adult Income Dataset.

Statistic	Value
Total records	48,842
Training/test records	32,561 / 16,281
Features	14 (6 numeric, 8 categorical)
At or below $50,000	37,155 (76.07%)
Above $50,000	11,687 (23.93%)
Male/female records	32,650 / 16,192
Male above-$50K rate	30.38%
Female above-$50K rate	10.93%
Missing categorical values	6,465

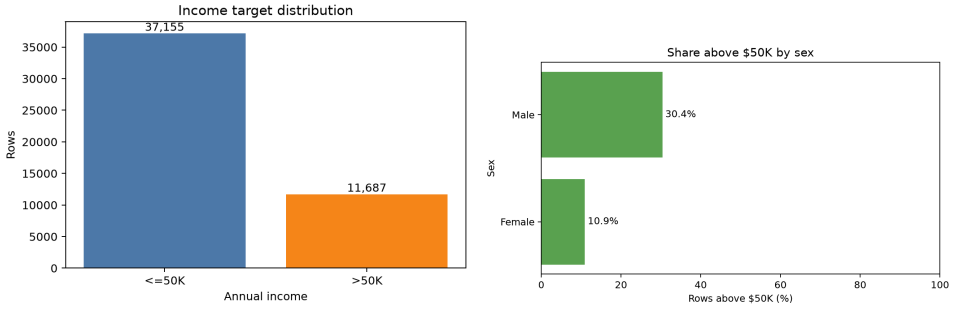


Figure 2: Income Distribution and Above-$50K Rates Across Sex Groups.

Ablate proxies and retrain model variants. We test these candidate proxies through controlled ablation. Individual features and selected feature sets are removed, and the XGBoost model is retrained under the same conditions. Comparing the resulting variants with the sex-excluded model shows whether proxy removal changes model performance, group disparities, or feature-level explanations.

Apply fairness-constrained learning. We compare feature removal with Fairlearn's `ExponentiatedGradient` reduction [Agarwal et al., 2018]. The constrained models use `sex` as the constraint variable and impose either demographic-parity or equalized-odds constraints, rather than using it as an ordinary model feature. This provides a direct comparison between proxy-focused feature interventions and fairness-constrained learning.

Common held-out evaluation. Finally, all model variants are evaluated on the same held-out data using a common prediction rule. The specific evaluation measures and their definitions are described in 4.3.

4.2 Dataset

We use the UCI Adult Income dataset [Dua and Graff, 2019], preserving its official training and test partitions. The dataset contains demographic, educational, occupational, and work-related variables, with a binary target indicating whether annual income exceeds $50,000. The main descriptive statistics are summarized in Table 1. The table records the dataset size, the official partition sizes, the six numeric and eight categorical features, the imbalanced target distribution, the observed `sex`-group sizes, and the number of categorical values encoded as `Missing`. Thus, it gives the scale and composition of the data used throughout the audit without presenting any model outcome.

We preprocess the data deterministically without dropping rows: whitespace is stripped, categorical `?` values are replaced with the explicit category `Missing`, numeric columns are converted to integers, and the income labels are mapped to 0 ($\leq$ $50,000) and 1 ($>$ $50,000). For modeling, the six numeric features pass through unchanged and the categorical features are one-hot encoded, with unknown test-time categories ignored. The resulting representation has 108 dimensions when `sex` is included and 106 dimensions when it is excluded; preprocessing is fitted on the training split only and `sex` is retained separately for group-level fairness evaluation.

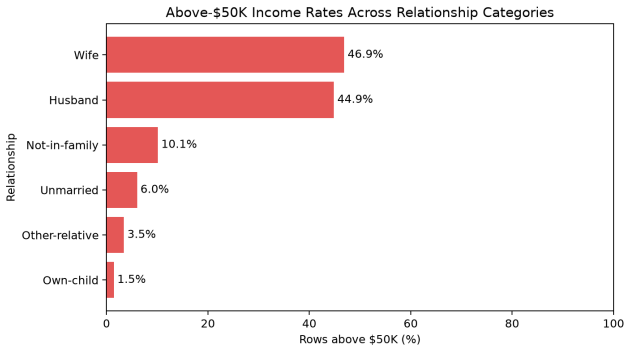


Figure 3: Above-$50K Income Rates Across Relationship Categories.

The target distribution and the observed group-specific rates are shown in Figure 2. The left panel shows that 11,687 of the 48,842 records (23.93%) are above $50,000, so overall accuracy alone could obscure performance on the less frequent class. The right panel shows that the above-$50K rate is 30.38% for males and 10.93% for females.

The `relationship` feature is examined descriptively in Figure 3 because it represents family and household structure and is one of the remaining variables that could retain group-related information. The observed above-$50K rates range from 1.46% for `Own-child` to 46.89% for `Wife`. These differences show that the feature has substantial structure relevant to the target, but they do not establish that it is a proxy for `sex`, identify a causal mechanism, or show that a trained model relies on it. We therefore use this plot as information and as a reference for the later TreeSHAP and ablation analyses. Together, the explicit sensitive attribute, the observed target disparity, and the presence of potentially related family and work-related features make UCI Adult Income suitable for a controlled proxy-feature audit, while the conclusions remain limited to the observed groups and dataset representation.

4.3 Evaluation metrics

We evaluate every model on the same held-out test split using predictive performance and group fairness metrics. Let $Y \in \{0, 1\}$ denote the true income label, $\widehat{Y}$ the binary prediction, and $A = \texttt{sex}$ the sensitive attribute. The observed groups (i.e., `Male` and `Female`) are indexed by $g \in \mathcal{G}$, and all metrics use the same fixed decision rule.

Predictive performance. We report accuracy (Acc), balanced accuracy (BAcc), precision (Pr), recall (Rc), and F1. Accuracy is the fraction of all test records classified correctly. Because the above-$50,000 class is less frequent, we also report balanced accuracy:

$$\mathrm{BAcc} = \frac{1}{2}\left(\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} + \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}\right).$$

BAcc gives equal weight to the two income classes and is therefore more informative than Acc when class frequencies are unequal. Precision is the fraction of positive predictions that are correct, Rc is the fraction of actual above-$50,000 records identified, and F1 is their harmonic mean. Together, these measures distinguish overall correctness from the model's ability to identify the minority income class and the reliability of its positive predictions.

Group fairness. We report demographic-parity difference (DPD) and equalized-odds difference (EOD) across the observed `sex` groups. As an auxiliary quantity for interpreting DPD and EOD, we also report the group-specific selection rate (also called positive prediction rate) as:

$$\mathrm{SR}_g = P(\widehat{Y} = 1 \mid A = g),$$

which is the proportion of records in group g receiving a positive prediction. DPD compares positive prediction rates without conditioning on the true income label:

$$\mathrm{DPD} = \max_{g \in \mathcal{G}} P(\widehat{Y} = 1 \mid A = g) - \min_{g \in \mathcal{G}} P(\widehat{Y} = 1 \mid A = g).$$

Table 2: Comparison of classifiers on the UCI Adult test split.

Model	Acc	BAcc	Pr	Rc	F1	DPD	EOD
Logistic Regression	0.7996	0.6131	0.7063	0.2595	0.3795	**0.0433**	0.0298
Decision Tree	0.8530	0.7509	0.7564	0.5572	0.6417	0.1367	0.0516
Random Forest	0.8512	0.7691	0.7158	0.6136	0.6608	0.1811	0.0842
HistGradientBoosting	0.8722	0.7947	0.7743	0.6477	0.7054	0.1765	0.0723
CatBoost	**0.8754**	0.7984	**0.7836**	0.6526	**0.7122**	0.1738	0.0762
Support Vector Machine	0.8016	0.6138	0.7251	0.2579	0.3805	0.0441	**0.0297**
KNN	0.7947	0.6264	0.6351	0.3073	0.4142	0.0771	0.0304
XGBoost	0.8746	**0.7990**	0.7786	**0.6557**	0.7119	0.1710	0.0604

This metric measures whether the model assigns the positive income classification at different rates across groups. DPD has range $[0, 1]$, where 0 indicates identical positive prediction rates, whereas larger values indicate greater disparity, with 1 being the maximum possible difference.

EOD measures disparities conditional on the true label:

$$\mathrm{EOD} = \max\left\{\max_{g\in\mathcal{G}} \mathrm{TPR}_g - \min_{g\in\mathcal{G}} \mathrm{TPR}_g,\ \max_{g\in\mathcal{G}} \mathrm{FPR}_g - \min_{g\in\mathcal{G}} \mathrm{FPR}_g\right\},$$

where $\mathrm{TPR}_g = P(\widehat{Y} = 1 \mid Y = 1, A = g)$ is the group-specific true-positive rate and $\mathrm{FPR}_g = P(\widehat{Y} = 1 \mid Y = 0, A = g)$ is the false-positive rate. EOD tells you how much a model's error rates differ across groups. It ranges from 0 to 1, with 0 indicating the model has comparable (fair) error rates across groups, whereas 1 suggests one group always gets the right outcome, while the other never.

5 Results

5.1 Classifier Selection

Before conducting the fairness audit, we compared eight classifiers on the same official UCI Adult training and held-out test partitions. Each model used all available features, including `sex`, together with the same training-only preprocessing. The comparison included linear, tree-based, distance-based, kernel, and boosting approaches. Table 2 reports the predictive and fairness results.

XGBoost achieved the highest balanced accuracy (0.7990) and recall (0.6557), while CatBoost achieved the highest accuracy (0.8754), precision (0.7836), and F1 score (0.7122). The difference between the two boosting models was small: CatBoost had slightly higher accuracy, precision, and F1, whereas XGBoost performed better on balanced accuracy and recall. The remaining models performed less strongly on the primary predictive measures, with the distance-based, linear, and kernel models showing particularly lower recall and F1 scores.

The fairness metrics show a similar trade-off rather than a universally dominant classifier. Logistic Regression and the Support Vector Machine had particularly low DPD (0.0433 and 0.0441) and EOD (0.0298 and 0.0297), indicating much smaller measured disparities than the boosting models. However, both models also had substantially weaker predictive performance, especially in recall and F1. Random Forest had the largest DPD and EOD, while KNN also traded lower disparities for weak predictive performance. XGBoost therefore provided the strongest overall predictive balance for the imbalanced classification task under the primary selection criteria and was selected for the subsequent sensitive-feature, proxy-feature, and fairness-mitigation analyses. This selection does not imply that XGBoost minimizes every fairness metric or has the highest value for every predictive metric.

5.2 XGBoost With and Without the Sensitive Feature

We first compared two XGBoost models on the same held-out test split: (1) a model using all features, including `sex`, and (2) a model excluding only `sex`. Table 3 reports the predictive performance of the two models. Their results are nearly identical: the sex-excluded model has Acc of 0.8748, BAcc of 0.7996, Pr of 0.7783, Rc of 0.6570, and F1 of 0.7125, compared with 0.8746, 0.7990, 0.7786, 0.6557,

Table 3: Predictive performance of XGBoost models trained with and without the sex.

Metric	With sex	Without sex	Difference
Acc	0.8746	0.8748	+0.0001
BAcc	0.7990	0.7996	+0.0005
Pr	0.7786	0.7783	-0.0004
Rc	0.6557	0.6570	+0.0013
F1	0.7119	0.7125	+0.0006

Table 4: Fairness metrics for XGBoost models trained with and without sex.

Model	Group	SR	DPD	EOD
With sex	Female	0.0849	0.1710	0.0604
With sex	Male	0.2559	–	–
Without sex	Female	0.0885	0.1662	0.0575
Without sex	Male	0.2548	–	–

and 0.7119, respectively, for the model that includes sex. Thus, excluding the explicit sensitive feature did not produce a meaningful change in the predictive performance observed on this test split.

The fairness results are shown in Table 4. The model including sex assigns positive predictions to 8.49% of female records and 25.59% of male records, yielding a DPD of 0.1710. After removing sex, the corresponding positive prediction rates are 8.85% and 25.48%, and DPD is 0.1662. The reduction in DPD is therefore small (0.0048 on the reported test split). EOD also changes only modestly, from 0.0604 to 0.0575. These results show two important observations: (1) removing sex helps reduce DPD and EOD, thereby improving the model's fairness, and (2) the small decrease suggests the presence of proxy-features that encode the information of sex, requiring further mitigation (i.e., removing sex alone was insufficient).

5.3 Proxy Identification and Removal

5.3.1 Proxy identification

We computed the mean absolute TreeSHAP value for each feature, representing its contribution to the prediction of the sex-excluded model. We also computed the association between a feature and sex to see if it encodes sex. In particular, we use Cramér's V for a categorical feature and absolute Spearman correlation for a numeric feature. The association value ranges from 0 to 1, with larger values indicating more reliance. We flagged a feature as a candidate proxy when it ranked in the upper half for both model reliance and sex association. Thus, an important predictive feature with weak sex association is not identified as a proxy, while a feature strongly associated with sex but weakly used by the model is not a high-priority model proxy.

Table 5 identifies marital-status, occupation, and hours-per-week as the features satisfying both ranking requirements. It also illustrates why both diagnostics are needed: relationship has the strongest sex association ($V_{\text{sex}} = 0.649$) but relatively low model reliance, whereas capital-gain has high model reliance but weak sex association ($V_{\text{sex}} = 0.067$). The result is a screening conclusion, not proof that any feature is causally a proxy. Figure 4 provides a visual summary of model reliance. marital-status is the most influential feature, while age and capital-gain are also influential but are not selected because their sex-association values in Table 5 are smaller.

5.3.2 Proxy removal

To assess whether the selected features contribute to the residual disparity, we retrained the sex-excluded model after removing each selected feature individually and after removing all three together. Thus, each proxy-ablation model excludes sex; for example, the model labelled Without marital-status excludes both sex and marital-status. The models labelled With sex and Without sex serve as the two reference models. Lower DPD and EOD indicate smaller observed disparities, while higher Acc, BAcc, Pr, Rc, and F1 indicate better predictive performance.

Table 5: Proxy identification using the sex-excluded model.

Feature	SHAP ↓	Association	Proxy?
marital-status	**1.276**	0.462	Yes
age	0.750	0.100	No
occupation	0.612	0.424	Yes
capital-gain	0.592	0.067	No
education-num	0.509	0.006	No
hours-per-week	0.394	0.265	Yes
relationship	0.292	**0.649**	No
capital-loss	0.183	0.042	No
workclass	0.145	0.153	No
fnlwgt	0.138	0.025	No
race	0.073	0.118	No
native-country	0.073	0.057	No
education	0.057	0.093	No

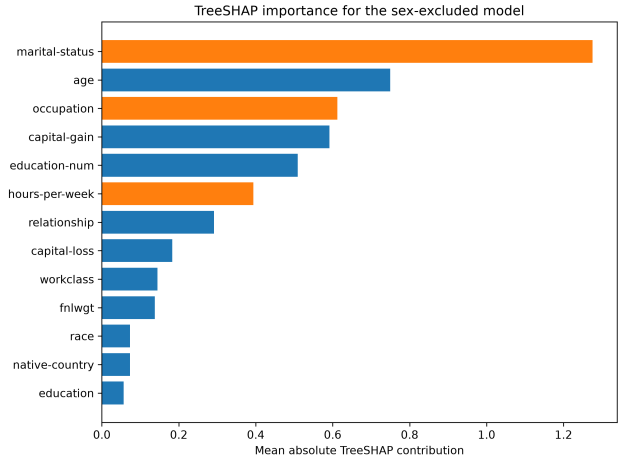


Figure 4: Mean absolute TreeSHAP contributions using the sex-excluded model. Orange bars indicate candidate proxies, while blue bars are not selected.

The individual ablations in Table 6 are modest and non-uniform. Removing `marital-status` barely changes DPD or EOD; removing `hours-per-week` slightly lowers both; and removing `occupation` causes the largest individual performance loss without a clearer fairness improvement. No single feature accounts for the residual disparity on its own. The joint removal is the clearest result. Relative to the sex-excluded model, DPD decreases from 0.1662 to 0.1576 and EOD from 0.0575 to 0.0528, while BAcc decreases from 0.7996 to 0.7844 and F1 from 0.7125 to 0.6923. The intervention therefore reduces, but does not eliminate, the disparity that remains after removing `sex`, and it does so at a measurable performance cost.

Figure 5 makes this result concrete: the sex-excluded model is close to the model using `sex`, while joint proxy removal moves toward lower DPD and lower BAcc. The analysis therefore supports a limited conclusion: removing `sex` is insufficient, and removing selected candidate proxies improves the measured disparity more than removing `sex` alone, but proxy deletion is neither cost-free nor complete. The candidates are empirical associations rather than proven causal mechanisms; interactions and the binary `sex` representation remain limitations.

5.4 Fairness-Constrained Retraining

We next evaluated whether fairness constraints could reduce the observed disparities more directly than feature removal. We trained two `ExponentiatedGradient` models using the sex-excluded feature set: one with a demographic-parity constraint and one with an equalized-odds constraint. In both cases, `sex` was used only as the sensitive feature for the constraint and evaluation, not as a

Table 6: Proxy ablation studies on predictive performance and fairness. Bold values are highest for performance metrics and lowest for fairness metrics.

Model	Acc	BAcc	Pr	Rc	F1	DPD	EOD
With sex	0.8746	0.7990	0.7786	0.6557	0.7119	0.1710	0.0604
Without sex	**0.8748**	**0.7996**	0.7783	**0.6570**	**0.7125**	0.1662	0.0575
Without marital-status	0.8736	0.7975	0.7763	0.6531	0.7094	0.1661	0.0580
Without occupation	0.8668	0.7858	0.7631	0.6323	0.6916	0.1638	0.0607
Without hours-per-week	0.8730	0.7943	**0.7792**	0.6451	0.7058	0.1657	0.0569
Without selected proxies	0.8690	0.7844	0.7775	0.6240	0.6923	**0.1576**	**0.0528**

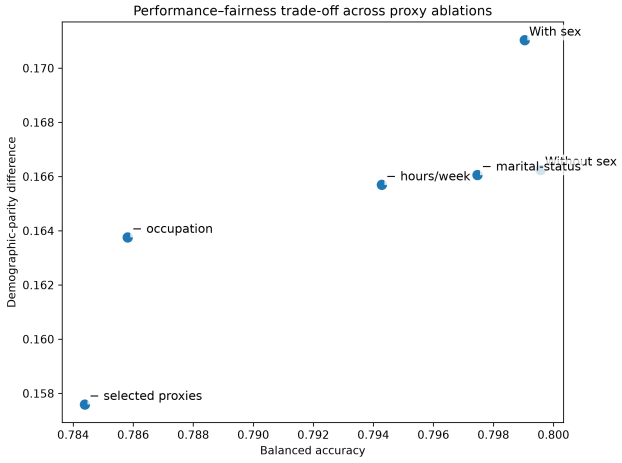


Figure 5: Balanced accuracy versus demographic parity difference trade-off.

predictive input. The models used the same fixed training and test partitions, preprocessing procedure, and held-out evaluation metrics as the unconstrained models.

Table 7 compares the constrained models with the sex-excluded model and the joint proxy-ablation model. The demographic-parity constraint reduced DPD from 0.1662 to 0.0122, substantially narrowing the difference in positive prediction rates. This improvement was accompanied by lower BAcc and F1, which decreased from 0.7996 and 0.7125 to 0.7405 and 0.6276, respectively. The equalized-odds constraint reduced EOD from 0.0575 to 0.0409, with a smaller performance reduction than demographic parity: BAcc was 0.7813 and F1 was 0.6904. However, its DPD remained 0.1241, showing that reducing conditional error-rate disparities does not produce equal selection rates.

The two constraints therefore produce different outcomes. Demographic parity nearly equalizes the overall positive prediction rates, but its EOD increases to 0.2904 because the resulting TPRs differ substantially between groups: 0.7763 for female records and 0.4859 for male records. Equalized odds instead reduces the TPR and FPR disparities, with TPRs of 0.6475 and 0.6066 and FPRs of 0.0335 and 0.0609 for female and male records, respectively. Its remaining DPD reflects the different group outcome prevalences in the test data. These results demonstrate that satisfying one operational fairness criterion does not guarantee the other and that constrained learning introduces a measurable performance–fairness trade-off.

Figure 6 shows the same trade-off in terms of balanced accuracy and the two fairness disparities. Figure 7 shows the group-level selection rates, TPRs, and FPRs underlying the aggregate DPD and EOD values. Together, the results indicate that fairness constraints provide more direct control over the targeted metric than proxy deletion, but they do not eliminate the need to state which fairness criterion is being optimized. In practice, the demographic-parity model is appropriate when similar positive prediction rates across groups are the primary institutional requirement, whereas the equalized-odds model is preferable when comparable error rates for qualified and unqualified individuals are more important.

Table 7: Predictive performance and fairness of unconstrained, proxy-ablated, and fairness-constrained models. Bold values are highest for performance metrics and lowest for fairness metrics.

Model	Acc	BAcc	Pr	Rc	F1	DPD	EOD
Without sex	**0.8748**	**0.7996**	0.7783	**0.6570**	**0.7125**	0.1662	0.0575
Without selected proxies	0.8690	0.7844	0.7775	0.6240	0.6923	0.1576	0.0528
Demographic parity	0.8513	0.7405	0.7684	0.5304	0.6276	**0.0122**	0.2904
Equalized odds	0.8702	0.7813	**0.7904**	0.6128	0.6904	0.1241	**0.0409**

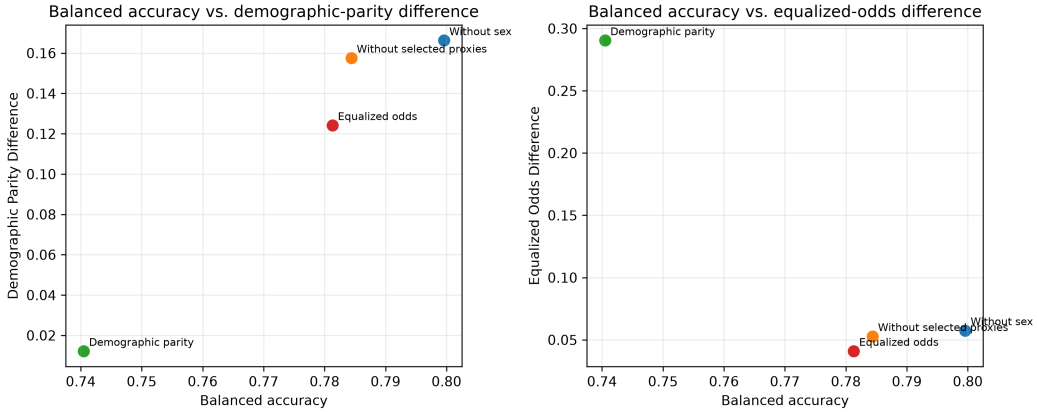


Figure 6: Performance–fairness trade-offs for the sex-excluded, proxy-ablated, and fairness-constrained models.

6 Discussion

Limitations. Although this study has shown satisfactory, two limitations remain. First, the proxy analysis is associational. That is, TreeSHAP and feature–`sex` association measure model reliance and statistical dependence, respectively, but do not establish causality. The selected ranking may also miss weaker or distributed proxy information. Second, DPD and EOD capture different properties and neither defines fairness in isolation; improving one may leave another unchanged or worsen it. The study therefore evaluates operational disparities under explicit definitions rather than certifying that any model or mitigation strategy is fair.

Future work. Future work should use causal and dependence-aware analyses to investigate distributed proxy information, compare multiple models and evaluation splits, and involve domain stakeholders in selecting fairness criteria and acceptable utility trade-offs for the intended application.

7 Conclusion

This study evaluated whether removing `sex` is sufficient to reduce fairness disparities in an income-classification model. In the UCI Adult experiment, excluding `sex` had almost no effect on predictive performance and only slightly reduced DPD and EOD, demonstrating that unawareness did not remove the observed group disparity. A TreeSHAP- and association-based screen identified candidate proxies, and joint removal of the selected features reduced both reported disparities further, but at a measurable cost in balanced accuracy and F1 and without eliminating disparity.

Fairness-constrained retraining produced stronger control of the targeted criterion, but the demographic-parity and equalized-odds models behaved differently across both fairness measures and predictive performance. The central conclusion is therefore not that one intervention is universally best. Rather, fairness claims require an explicit criterion, held-out evaluation of both utility and group outcomes, and analysis of information that remains after the sensitive feature is removed. For this dataset and experimental design, removing `sex` is a useful comparison condition, not a sufficient fairness intervention. These conclusions are limited by the the associational nature of the proxy

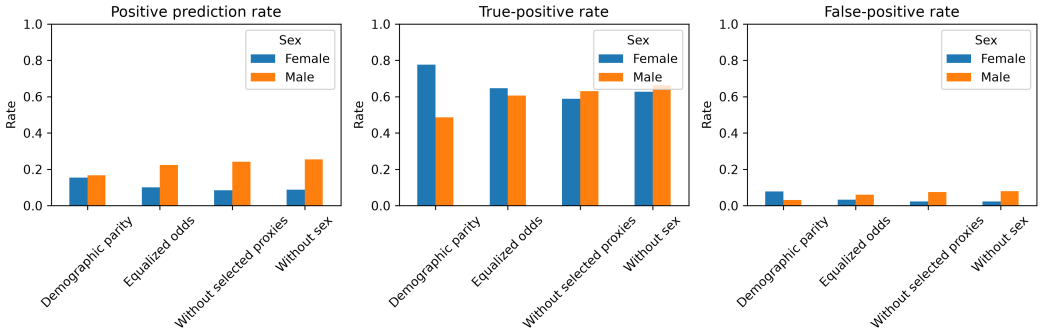


Figure 7: Female and male selection rates, true-positive rates, and false-positive rates for the compared model variants.

analysis and the differing meanings of the fairness metrics. Future work should investigate proxy relationships causally and select fairness criteria with stakeholders whose decisions and harms are represented by the application.

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